



Practice Assignment 3

Instruction:

* Students are allowed to write their answers (like SQL queries, Screen shot of outputs, etc.) in word file (Answer sheet) provided by instructor. After finishing the assignment, students must convert the word file (Answer sheet) into a PDF file. Finally, students upload the file in Moodle.

1. Create the following tables in a new database 'SaleManagerment':

- Create a new database with the name " SaleManagerment".
- Create all the tables under this database.

2. Informations Tables:

a) Table Name: **Clients**

Column Name	Data Type	Size	Constraints	
Client_Number	VARCHAR	10	Primary Key, first letter must start with 'C'	<pre>create table clients(Client_Number varchar(10), Client_Name varchar(25) not null, Address varchar(30), City varchar(30), Pincode int not null, Province char(25), Amount_Paid decimal(15,4), Amount_Due decimal(15,4), check(Client_Number like 'C%'), primary key(Client_Number));</pre>
Client_Name	VARCHAR	25	Not null	
Address	VARCHAR	30		
City	VARCHAR	30		
Pincode	INT		Not null	
Province	CHAR	25		
Amount_Paid	DECIMAL	15,4		
Amount_Due	DECIMAL	15,4		

a) Table Name: **Product**

Column Name	Data Type	Size	Constraints	
Product_Number	VARCHAR	15	Primary Key, first letter must start with 'P'	<pre>create table product(Product_Number varchar(15), Product_Name varchar(25) not null unique, Quantity_On_Hand int not null, Quantity_Sell int not null, Sell_Price decimal(15,4) not null, Cost_Price decimal(15,4) not null, check(Product_Number like 'P%'), check(Cost_Price > 0), primary key(Product_Number));</pre>
Product_Name	VARCHAR	25	Not null, Unique	
Quantity_On_Hand	INT		Not null	
Quantity_Sell	INT		Not null	
Sell_Price	DECIMAL	15,4	Not null	
Cost_Price	DECIMAL	15,4	Not null, cannot be 0	

b) Table Name: Salesman

Column Name	Data Type	Size	Constraints	<pre> create table Salesman(Salesman_Number varchar(15), Salesman_Name varchar(25) not null, Address varchar(30), City varchar(30), Pincode int not null, Province char(25) default('Viet Nam'), Salary decimal(15,4) not null, Sales_Target int not null, Target_Achieved int, Phone char(10) not null unique, check(Salesman_Number like 'S%'), check(Salary > 0), check(Sales_Target > 0), primary key(Salesman_Number)); </pre>
Salesman_Number	VARCHAR	15	Primary Key, first letter must start with 'S'	
Salesman_Name	VARCHAR	25	Not null	
Address	VARCHAR	30		
City	VARCHAR	30		
Pincode	INT		Not null	
Province	CHAR	25	Default 'Vietnam'	
Salary	DECIMAL	15,4	Not null, cannot be 0	
Sales_Target	INT		Not null, cannot be 0	
Target_Achieved	INT			
Phone	CHAR	10	Not null, Unique,	

c) Table Name: SalesOrder

Column Name	Data Type	Size	Constraints
Order_Number	VARCHAR	15	Primary Key, first letter must start with 'O'
Order_Date	DATE		
Client_Number	VARCHAR	15	Foreign key references Client_Number of Client table, first letter must start with 'C'
Salesman_Number	VARCHAR	15	Foreign key references Salesman_Number of Salesman table, first letter must start with 'S'
Delivery_Status	CHAR	15	Values ('Delivered', 'On Way', 'Ready to Ship')
Delivery_Date	DATE		Cannot be less than Order_date
Order_Status	VARCHAR	15	Values ('In Process', 'Successful', 'Cancelled')

```

create table SalesOrder(
    Order_Number varchar(15),
    Order_Date date,
    Client_Number varchar(15),
    Salesman_Number varchar(15),
    Delivery_Status char(15),
    Delivery_Date date,
    Order_Status varchar(15),
    primary key (Order_Number),
    foreign key (Client_Number) references clients(Client_Number),
    foreign key (Salesman_Number) references salesman(Salesman_Number),
    check (Order_Number like 'O%'),
    check (Client_Number like 'C%'),
    check (Salesman_Number like 'S%'),
    check (Delivery_Status in ('Delivered', 'On Way', 'Ready to Ship')),
    check (Delivery_Date > Order_Date),
    check (Order_Status in ('In Process', 'Successful', 'Cancelled'))
);

```

d) Table Name: SalesOrderDetails

Column Name	Data Type	Size	Constraints	<pre> create table SalesOrderDetails(Order_Number varchar(15), Product_Number varchar(15), Order_Quantity int, Discount_Rate int, check (order_number like '0%'), check (Product_Number like 'P%'), foreign key (Order_Number) references salesorder(Order_Number), foreign key (Product_Number) references product (Product_Number)); </pre>
Order_Number	VARCHAR	15	Foreign key references Order_Number of Sales_Order table	
Product_Number	VARCHAR	15	Foreign key references Product_Number of Product table, first letter must start with 'P'	
Order_Quantity	INT			

3. Insert Data into following tables:

a) Clients

Client_Number	Client_Name	Address	City	Pincode	Province	Amount_Paid	Amount_Due
C101	Mai Xuan	Phu Hoa	Dai An	700001	Binh Duong	10000	5000
C102	Le Xuan	Phu Hoa	Thu Dau Mot	700051	Binh Duong	18000	3000
C103	Trinh Huu	Phu Loi	Da Lat	700051	Lam Dong	7000	3200
C104	Tran Tuan	Phu Tan	Thu Dau Mot	700080	Binh Duong	8000	0
C105	Ho Nhu	Chanh My	Hanoi	700005	Hanoi	7000	150
C106	Tran Hai	Phu Hoa	Ho Chi Minh	700002	Ho Chi Minh	7000	1300
C107	Nguyen Thanh	Hoa Phu	Dai An	700023	Binh Duong	8500	7500
C108	Nguyen Sy	Tan An	Da Lat	700032	Lam Dong	15000	1000
C109	Duong Thanh	Phu Hoa	Ho Chi Minh	700011	Ho Chi Minh	12000	8000
C110	Tran Minh	Phu My	Hanoi	700005	Hanoi	9000	1000

Script to insert data:

```

('C101','Mai Xuan','Phu Hoa','Dai An',700001,'Binh Duong',10000,5000),
('C102','Le Xuan','Phu Hoa','Thu Dau Mot',700051,'Binh Duong',18000,3000),
('C103','Trinh Huu','Phu Loi','Da Lat',700051,'Lam Dong ',7000,3200),
('C104','Tran Tuan','Phu Tan','Thu Dau Mot',700080,'Binh Duong',8000,0),
('C105','Ho Nhu','Chanh My','Hanoi',700005,'Hanoi',7000,150),
('C106','Tran Hai','Phu Hoa','Ho Chi Minh',700002,'Ho Chi Minh',7000,1300),
('C107','Nguyen Thanh ','Hoa Phu','Dai An',700023,'Binh Duong',8500,7500),
('C108','Nguyen Sy','Tan An','Da Lat',700032,'Lam Dong ',15000,1000),
('C109','Duong Thanh','Phu Hoa','Ho Chi Minh',700011,'Ho Chi Minh',12000,8000),
('C110','Tran Minh','Phu My','Hanoi',700005,'Hanoi',9000,1000);

```

b) Product

Product_Number	Product_Name	Quantity_On_Hand	Quantity_Sell	Sell_Price	Cost_Price
P1001	TV	10	30	1000	800
P1002	Laptop	12	25	1500	1100
P1003	AC	23	10	400	300
P1004	Modem	22	16	250	230
P1005	Pen	19	13	12	8
P1006	Mouse	5	10	100	105
P1007	Keyboard	45	60	120	90
P1008	Headset	63	75	50	40

Script to insert data:

```
( 'P1001','TV',10,30,1000,800),
( 'P1002','Laptop',12,25,1500,1100),
( 'P1003','AC',23,10,400,300),
( 'P1004','Modem',22,16,250,230),
( 'P1005','Pen',19,13,12,8),
( 'P1006','Mouse',5,10,100,105),
( 'P1007','Keyboard',45,60,120,90),
( 'P1008','Headset',63,75,50,40);
```

c) Salesman

Salesman_Number	Salesman_Name	Address	City	Pincode	Province	Salary	Sales_Target	Target_Achieve	Phone
S001	Huu	Phu Tan	Ho Chi Minh	700002	Ho Chi Minh	15000	50	35	0902361123
S002	Phat	Tan An	Hanoi	700005	Hanoi	25000	100	110	0903216542
S003	Khoa	Phu Hoa	Thu Dau Mot	700051	Binh Duong	17500	40	30	0904589632
S004	Tien	Phu Hoa	Dai An	700023	Binh Duong	16500	70	72	0908654723
S005	Deb	Hoa Phu	Thu Dau Mot	700051	Binh Duong	13500	60	48	0903213659
S006	Tin	Chanh My	Da Lat	700032	Lam Dong	20000	80	55	0907853497

Script to insert data:

```
( 'S001','Huu','Phu Tan','Ho Chi Minh',700002,'Ho Chi Minh',15000,50,35,'0902361123'),
( 'S002','Phat','Tan An','Hanoi',700005,'Hanoi',25000,100,110,'0903216542'),
( 'S003','Khoa','Phu Hoa','Thu Dau Mot',700051,'Binh Duong',17500,40,30,'0904589632'),
( 'S004','Tien','Phu Hoa','Dai An',700023,'Binh Duong',16500,70,72,'0908654723'),
( 'S005','Deb','Hoa Phu','Thu Dau Mot',700051,'Binh Duong',13500,60,48,'0903213659'),
( 'S006','Tin','Chanh My','Da Lat',700032,'Lam Dong',20000,80,55,'0907853497');
```

d) Sales_order

Order_Number	Order_Date	Client_Number	Salesman_Number	Delivery_Status	Delivery_Date	Order_Status
O20001	2022-01-15	C101	S003	Delivered	2022-02-10	Successful
O20002	2022-01-25	C102	S003	Delivered	2022-02-15	Cancelled
O20003	2022-01-31	C103	S002	Delivered	2022-04-03	Successful
O20004	2022-02-10	C104	S003	Delivered	2022-04-23	Successful
O20005	2022-02-18	C101	S003	On Way	null	Cancelled
O20006	2022-02-22	C105	S005	Ready to Ship	null	In Process
O20007	2022-04-03	C106	S001	Delivered	2022-05-08	Successful
O20008	2022-04-16	C102	S006	Ready to Ship	null	In Process
O20009	2022-04-24	C101	S004	On Way	null	Successful
O20010	2022-04-29	C106	S006	Delivered	2022-05-08	Successful
O20011	2022-05-08	C107	S005	Ready to Ship	null	Cancelled
O20012	2022-05-12	C108	S004	On Way	null	Successful
O20013	2022-05-16	C109	S001	Ready to Ship	null	In Process
O20014	2022-05-16	C110	S001	On Way	null	Successful

Script to insert data:

```
( 'O20001','2022-01-15','C101','S003','Delivered','2022-02-10','Successful'),
( 'O20002','2022-01-25','C102','S003','Delivered','2022-02-15','Cancelled'),
( 'O20003','2022-01-31','C103','S002','Delivered','2022-04-03','Successful'),
( 'O20004','2022-02-10','C104','S003','Delivered','2022-04-23','Successful'),
( 'O20005','2022-02-18','C101','S003','On Way',null,'Cancelled'),
( 'O20006','2022-02-22','C105','S005','Ready to Ship',null,'In Process'),
( 'O20007','2022-04-03','C106','S001','Delivered','2022-05-08','Successful'),
( 'O20008','2022-04-16','C102','S006','Ready to Ship',null,'In Process'),
( 'O20009','2022-04-24','C101','S004','On Way',null,'Successful'),
( 'O20010','2022-04-29','C106','S006','Delivered','2022-05-08','Successful'),
( 'O20011','2022-05-08','C107','S005','Ready to Ship',null,'Cancelled'),
( 'O20012','2022-05-12','C108','S004','On Way',null,'Successful'),
( 'O20013','2022-05-16','C109','S001','Ready to Ship',null,'In Process'),
( 'O20014','2022-05-16','C110','S001','On Way',null,'Successful');
```

e) Sales_Order_Details

Order_Number	Product_Number	Order_Quantity	Script to insert data:
O20001	P1001	5	
O20001	P1002	4	
O20002	P1007	10	
O20003	P1003	12	

O20004	P1004	3	('O20004','P1004',3), (('O20005','P1001',8), (('O20005','P1008',15), (('O20005','P1002',14), (('O20006','P1002',5), (('O20007','P1005',6), (('O20008','P1004',8), (('O20009','P1008',2), (('O20010','P1006',11), (('O20010','P1001',9), (('O20011','P1007',6), (('O20012','P1005',3), (('O20012','P1001',2), (('O20013','P1006',10), (('O20014','P1002',20);
O20005	P1001	8	
O20005	P1008	15	
O20005	P1002	14	
O20006	P1002	5	
O20007	P1005	6	
O20008	P1004	8	
O20009	P1008	2	
O20010	P1006	11	
O20010	P1001	9	
O20011	P1007	6	
O20012	P1005	3	
O20012	P1001	2	
O20013	P1006	10	
O20014	P1002	20	

4. Write SQL queries for following operations:

- Show the schema of “Clients”, “Product”, “Salesman”, “Salesorder”, “Salesorderdetails” tables.
- Display the content of “Clients”, “Product”, “Salesman”, “Salesorder”, “Salesorderdetails” tables.

Practice in Class

a) Using Logical Operators (AND, OR, NOT, LIKE, IN, BETWEEN) and Comparison Operators (=, >, <, >=, <=, <>)

1. Show the all-clients details who lives in “Binh Duong”.
2. Find the client’s number and client’s name who do not live in “Hanoi”.
3. Identify the names of all products with less than 25 in stock.
4. Find the product names where company making losses.
5. Find the salesman’s details who are able achieved their target.
6. Select the names and city of salesman who are not received salary between 10000 and 17000.
7. Show order date and the clients numbers who are bought the product between '2022-01-01' and '2022-02-15'.

b) Finding strings in table

8. Find the names of cities in clients table where city name starts with "N"
9. Display clients’ information whose names have "u" in third position.

10. Find the details of clients whose names have "u" in second last position.
11. Find the names of cities in clients table where city name starts with "D" and ends with "n".
12. Select all clients details who belongs from Ho Chi Minh, Hanoi and Da Lat.
13. Choose all clients data who do not reside in Ho Chi Minh, Hanoi and Da Lat.

c) Using mySQL functions (Min(), Max(), COUNT(), AVG() and SUM())

14. Find the average salesman's salary.
15. Find the name of the highest paid salesman.
16. Find the name of the salesman who is paid the lowest salary.
17. Determine the total number of salespeople employed by the company.
18. Compute the total salary paid to the company's salesman.

d) Using Order by keyword, limit clause

19. Select the salesman's details sorted by their salary.
20. Display salesman names and phone numbers based on their target achieved (in ascending order) and their city (in descending order).
21. Display 3 first names of the salesman table and the salesman's names in descending order.
22. Find salary and the salesman's names who is getting the highest salary.
23. Find salary and the salesman's names who is getting second lowest salary.
24. Display the first five sales order information from the sales order table.
25. Display next ten sales order information from sales order table except first five sales order.

e) Using group by with having clause with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()), order by.

Having syntax:

```
SELECT column_name(s)
FROM table_name
WHERE condition
GROUP BY column_name(s)
HAVING condition
ORDER BY column_name(s);
```

26. If there are more than one client, find the name of the province and the number of clients in each province, ordered high to low.
27. Display information clients have number of sales order more than 1.

Practice to grade

Question : Using database 'SaleManagerment' to write SQL queries following:

28. Display the name and due amount of those clients who lives in "Hanoi".
29. Find the clients details who has due more than 3000.
30. Find the clients name and their province who has no due.
31. Show details of all clients paying between 10,000 and 13,000.
32. Find the details of clients whose name is "Dat".
33. Display all product name and their corresponding selling price.
34. How many TVs are in stock?
35. Find the salesman's details who are not able achieved their target.
36. Show all the product details of product number 'P1005'.
37. Find the buying price and sell price of a Mouse.
38. Find the salesman names and phone numbers who lives in Thu Dau Mot.
39. Find all the salesman's name and salary.
40. Select the names and salary of salesman who are received between 10000 and 17000.
41. Display all salesman details who are received salary between 10000 and 20000 and achieved their target.
42. Display all salesman details who are received salary between 20000 and 30000 and not achieved their target.
43. Find information about all clients whose names do not end with "h".
44. Find client names whose second letter is 'r' or second last letter 'a'.
45. Select all clients where the city name starts with "D" and at least 3 characters in length.
46. Select the salesman name, salaries and target achieved sorted by their target_achieved in descending order.
47. Select the salesman's details sorted by their sales_target and target_achieved in ascending order.
48. Select the salesman's details sorted ascending by their salary and descending by achieved target.
49. Display salesman names and phone numbers in descending order based on their sales target.
50. Display the product name, cost price, and sell price sorted by quantity in hand.
51. Retrieve the clients' names in ascending order.
52. Display client information based on order by their city.
53. Display client information based on order by their address and city.
54. Display client information based on their city, sorted high to low based on amount due.

55. Display the data of sales orders depending on their delivery status from the current date to the old date.
56. Display last five sales order information from sales order table.
57. Count the pincode in client table.
58. How many clients are living in Binh Duong?
59. Count the clients for each province.
60. If there are more than three clients, find the name of the province and the number of clients in each province.
61. Display product number and product name and count number orders of each product more than 1 (in ascending order).